

Multi-Source Telemetry Console

Preserve health evidence across presentation and records

A deterministic, receive-only project with no actuator

Lesson	027 — project	Time	240–330 minutes
Prerequisites	Lessons 001–026	Board	Arduino Mega 2560
Evidence	RGB cadence and D13	Status	host verification; bench open
Energy class	E1: USB, passive buttons, resistor-limited LEDs		

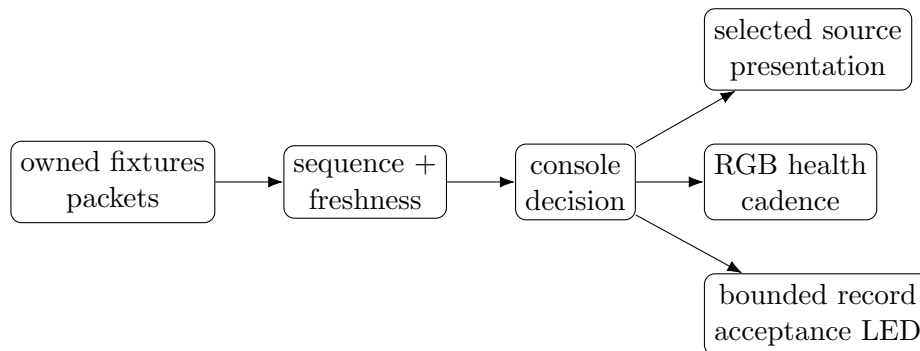
Boundary. This is an educational annunciator for owned lab fixtures. It is not a fire, burglar, medical, industrial, environmental-safety, or emergency system. It does not actuate anything, dispatch help, transmit, replay, clone, or interpret an unknown protocol.

1 Mission and evidence chain

Compose the receive-only arc without losing the distinction between evidence and presentation:

source evidence → normalized reading → console decision → visible state → stable record.

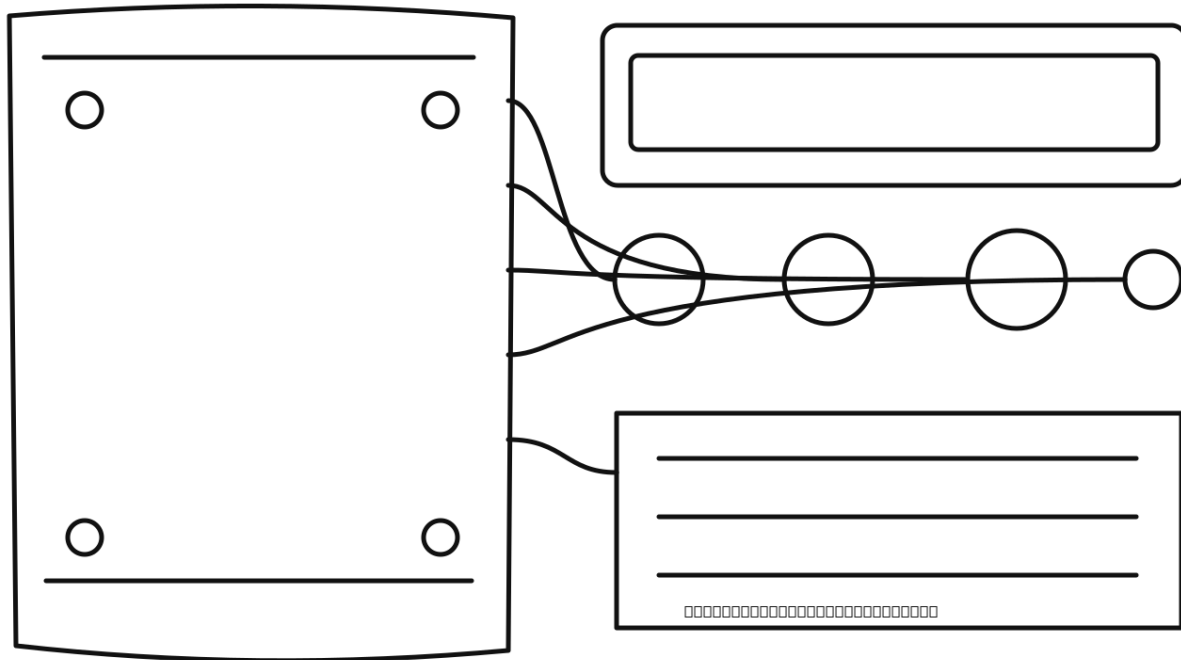
The released exercise uses three compiled-in, owned fixture observations: temperature, relative humidity, and contact. A live receiver is not required. Input order cannot change configured source order. Missing sources remain explicit. Presentation and record failures cannot rewrite health.



1.1 Learning outcomes

- distinguish capture, frame, packet, record, and presentation;
- explain configured slot order and explicit missing observations;
- derive health from quality, sequence condition, and freshness;
- keep acknowledgement separate from health;
- prove record acceptance without confusing it with source validity;
- replay an exact mixed-source trace with byte-identical output.

2 Orientation and circuit plan



Alternative text: graphite orientation plate showing a Mega, optional character display, next and acknowledge buttons, RGB source-health evidence, record LED, and separate paths from console decision to presentation and record acceptance. No radio or actuator is shown.

The plate shows the extension path as well as the starter circuit. The released sketch's pin table below is authoritative; the display block is optional further work and owns no pin in the starter circuit.

Circuit element	Released role	Electrical constraint
Next button, D30	advance configured source slot	internal pull-up; switch to GND
Acknowledge button, D31	acknowledge current attention intent	internal pull-up; switch to GND
RGB LED, D5–D7	health plus non-color cadence	one 330 ohm resistor per channel
D13 LED	toggles after complete record acceptance	built-in LED, independent of health
Source fixtures	deterministic observation evidence	compiled-in, owned lab values

Unpowered prediction. Circle every resource owner in the final sketch. For each borrower, draw an arrow to the owner that must outlive it. Confirm button wiring, one resistor per RGB channel, common ground, and no display, sounder, relay, external load supply, transmitter, or mains connection.

3 Vocabulary worksheet

Word	Meaning in this arc	What it does not mean
Capture	timestamped physical evidence	decoded command
Frame	bounded protocol candidate	trusted packet
Packet	ADK-owned telemetry envelope	authenticated identity
Record	stable storage representation	source truth
Presentation	human-visible interpretation	mutation of observation health

CRC detects accidental corruption. It does not authenticate a sender, hide content, prove a location, or prevent replay. Source identifiers organize owned fixtures; they are not credentials. Freshness bounds local age; it does not prove a remote clock.

4 The narrative loop

```
observeTelemetrySources (now);
decideConsoleState      (now);
presentConsoleState     (now);
recordConsoleEvidence   (now);
```

observe builds the fixed observation fixture. **decide** submits one complete configured-slot view through **TelemetryConsoleProject**. **present** checks the RGB presentation result. **record** checks the bounded record result. No stage hides time or allocates memory.

4.1 Object-order exercise

Number these in construction order: ____ console, ____ runtime, ____ project, ____ fixture observations, ____ buttons, ____ record sink, ____ RGB presentation.

Then answer:

1. Which object decides source health? _____
2. Which object decides whether one stable record is pending? _____
3. Which failure may silence a cue without changing health? _____
4. Why is packet arrival order excluded from source selection? _____

5 Health precedence

Health	Required interpretation	Evidence
Starting	expected slots have not all produced usable observations	blue cadence
Healthy	every expected slot is present, valid quality, and fresh	green cadence
Degraded	an observation is aging or reports a sequence gap	amber cadence
Fault	stale, invalid, corrupt, duplicate, or missing evidence	fast red cadence
Stopped	lifecycle inactive	unlit

Fault dominates degraded; degraded dominates healthy; startup is not silently healthy. When several conditions coexist, the same input snapshot must always choose the same result.

5.1 Precedence worksheet

Input set	Prediction	Observation	Interpretation
all valid/fresh	_____	_____	_____
one aging + one gap	_____	_____	_____
one stale + one aging	_____	_____	_____
duplicate identity	_____	_____	_____
missing configured slot	_____	_____	_____

6 Operator intent is not evidence

The next action selects a configured slot, not a source ID or packet-arrival position. A held button produces one debounced event. Selection wraps after the last configured slot.

Acknowledgement applies only to the current attention intent. The starter circuit has no sounder; inspect the state in host replay, then add a bounded sounder only as the extension exercise. Acknowledgement does not:

- convert fault to healthy;
- clear a missing or stale source;
- erase or rewrite a record;
- prevent a new fault from being announced.

Experiment. Enter fault, acknowledge it, and keep the fault present. Predict fast red evidence will remain. Inspect the replay snapshot to see **Attention** become **Notice**. Recover, then introduce a new fault and predict **Attention** again.

7 Presentation and record separation

The released presentation converts health to an RGB color and a distinct cadence. Its selected-source argument stays available for a later display without entering the decision. A presentation failure leaves the source snapshot unchanged.

The record policy writes on accepted observation, health transition, acknowledgement, or periodic heartbeat. Simultaneous reasons become one record under fixed priority. A failed write remains visible and may retry only under a bounded deterministic policy. No queue grows in SRAM.

Failure	Must remain unchanged	Separate evidence
presentation write	source health and pending record	returned status
record append	source health and selected slot	D13 does not toggle
record retry	accepted observation snapshot	same bounded record bytes
attention extension	health and record intent	red evidence remains

7.1 Golden-record worksheet

Copy one released canonical record exactly:

Trigger reason _____
 Caller capacity _____ bytes
 Encoded size _____ bytes
 First differing byte after corruption _____
 Retry byte-identical? yes / no; evidence: _____

8 Deterministic replay lab

Prepare one exact trace:

1. start with all three slots missing;
2. recover temperature, humidity, then contact in configured order;
3. deliver later packets in a different arrival order;
4. age one source to the exact degraded boundary;
5. cross the stale boundary;
6. acknowledge attention while fault remains;
7. recover, then introduce a different fault;
8. inject one record failure and bounded retry;
9. shut down from the active fault state.

Time	Source change	Expected health	Selected slot	Record reason
_____	all missing	_____	_____	_____
_____	first recovery	_____	_____	_____
_____	all recovered	_____	_____	_____
_____	exact aging	_____	_____	_____
_____	exact stale	_____	_____	_____
_____	acknowledge	_____	_____	_____

Replay the same timestamps, packet bytes, and button events twice. Compare health, signal intent, selected slot, record reasons, and record bytes. Every difference is a failure until explained.

9 Predict, observe, interpret

9.1 Resource acquisition

Predict: blue cadence appears only after all required owners initialize. **Observe:** startup cadence without Serial. **Interpret:** this supports resource acquisition; it does not prove shutdown release or a valid source.

9.2 Source health

Predict: the trace advances blue, green, amber, red. **Observe:** the RGB cadence at TP-R, TP-G, and TP-B. **Interpret:** agreement supports the console decision. It does not authenticate a fixture or establish sensor accuracy.

9.3 Record acceptance

Predict: D13 toggles only after a complete record is accepted. **Observe:** inject a failed append in the host fixture and compare the acceptance state with health evidence. **Interpret:** D13 proves the simulated storage

call path, not physical media durability.

9.4 Safe shutdown

Predict: RGB and record evidence become inactive and owned pins release. **Observe:** measure D5–D7 and D13 against GND after the stop path. **Interpret:** inactive LEDs alone do not prove high impedance or claim release.

10 Diagnosis

Observation	Check in order
stays Starting	configured count, explicit presence, first observation time
Fault instead of Degraded	stale boundary, invalid quality, duplicate ID, missing slot
selection follows arrival	configured-slot table; never sort by packet arrival
acknowledgement clears red	separate sound acknowledgement from health state
D13 toggles on failed record	acceptance must follow a successful append
new fault lacks attention	fault-episode transition and acknowledgement reset
different second replay	hidden clock, mutable fixture, uninitialized padding, ordering

11 CLI workflow and acceptance

```
make host-test
make host-test-sanitize
make telemetry-fixtures-check
make arduino-Lesson027TelemetryConsole
make size-Lesson027TelemetryConsole
make lessons
make lessons-check
make site-check
make upload EXAMPLE=Lesson027TelemetryConsole PORT=/dev/ttyACM0
make hardware-card LESSON=027 BOARD=mega2560
```

These targets verify code and publication, not the open physical card. There is intentionally no transmit, replay, clone, brute-force, scan, or protocol-export target.

11.1 Unsigned physical acceptance card

- ☐ Exact Mega revision, buttons, LEDs, supply, and instrument recorded.
- ☐ Wiring and resistor values checked with USB removed.
- ☐ Startup acquisition evidence observed.
- ☐ Healthy, degraded, fault, acknowledgement, reannouncement, and recovery evidence observed.
- ☐ Record failure kept separate from source health.
- ☐ Shutdown output state measured at named test points.
- ☐ No transmitter, actuator, relay, external load, or mains circuit present.

Board and revision	_____
Exact specimens and markings	_____
Supply and instruments	_____
Firmware commit and tool versions	_____
Predictions	_____
Observations	_____
Interpretation and deviations	_____
Operator and date	_____

Open gate. Host replay, firmware compilation, a generated PDF, and visual inspection are not physical verification. Leave this card blank until the released circuit is measured on a Mega 2560.

12 Further work

1. Add a fourth generated source without changing arrival-order behavior.
2. Design an explicit log-full policy with no heap and no silent overwrite.
3. Explain what cryptographic authentication would require and why CRC, source ID, sequence, and freshness do not provide it.
4. Add a bounded sounder for attention while retaining the RGB state as equivalent non-audible evidence.
5. Audit the final object graph for ownership, rollback, shutdown order, and exception-unwinding safety in a caller.

Package status

The interface, deterministic host tests, canonical Mega sketch, and printable package are present. Physical Mega acceptance remains open. The exercise makes no live-radio or multi-room hardware claim.